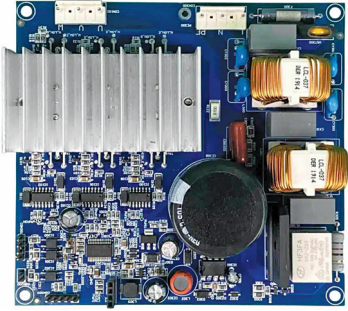


The author, Nidhi Agarwal, works as Technology Journalist at EFY

Refrigerator Compressor with BLDC Sensorless Control	Refrigerator Compressor Driver	Refrigerator Compressor
		
The refrigerator compressor reference design simplifies refrigerator functionality by integrating compressor and system controls on a single board, enhancing efficiency and reducing complexity. NXP Semiconductors' refrigerator compressor reference design features a brushless DC motor (BLDC) with sensorless control, achieving greater efficiency through a field-oriented control (FOC) based on the MC56F82xxx controller. The motor drive board is designed to optimise efficiency and performance, with a low standby power consumption of less than 0.6W. It can handle an input voltage range from 80% to 110% of the rated mains voltage. The system is equipped with a high-performance VF refrigerator compressor control solution that meets Class 1 energy efficiency standards, incorporating a sensorless FOC with closed-loop control of both speed and current. Additionally, the controller is IEC60730 certified, complying with international safety standards. Typically, refrigerators have two control segments: compressor and system control. System control monitors compartment temperatures to regulate compressor speed, fan states, and defrost heater operation, and manages a control panel. It uses PWM signals to set compressor speeds from 1200 to 4500rpm. Traditionally, these functions are handled by separate microcontroller units for system, compressor, and human-machine interface (HMI). However, integrating both controls onto a single board could simplify the design, eliminating the need for separate AC-DC power circuits. The reference design addresses the growing trend of using BLDC motors in reciprocating refrigerator compressors. NXP Semiconductor https://www.electronicsforu.com/electronics-projects/refrigerator-compressor-reference-design-with-blcd-sensorless-control	The reference design with an input range of 165 to 265Vrms includes a field oriented control (FOC) solution, power factor correction (PFC) control, and a graphical interface (GUI), offering EMI performance and ease of operation. The REF_Fridge_C101T_6ED from Infineon Technologies, tailored for refrigerator compressors, is a key part of the iMOTION series for motor drives. It operates within a 165 to 265Vrms input voltage range, with motor power output decreasing at lower AC inputs. At 220V AC and 25°C ambient temperature, it draws a maximum input current of 1.5Arms. Under the same voltage and temperature, but with a 5kHz PWM frequency and 100°C case temperature without a heat sink, it has a maximum input power of 240W and motor current of 0.66Arms. The device accommodates a maximum DC bus voltage of 400V and features a shunt resistance of 150mΩ. This design features key components like the 6EDL04I06PT EiceDRIVER Compact, a 600V full-bridge three-phase gate driver IC, and the iMOTION IMC101T-T038 for both sensorless and Hall-sensor motor drives. The design offers multiple benefits, such as a field-proven, turnkey iMOTION sensorless field oriented control (FOC) solution, optional power factor correction (PFC) control, optimisation for full-load operation and electromagnetic interference (EMI) performance, and a user-friendly graphical interface (GUI) for simplified operation. The reference design is particularly well-suited for refrigerator compressors. Infineon Technologies https://www.electronicsforu.com/electronics-projects/reference-design-of-refrigerator-compressor-driver	The refrigerator compressor reference design offers sensorless control and cost-effective single-shunt current sensing. It handles up to 350W in high-pressure situations, and is designed for reliable, low-power operation. Microchip's reference design for a refrigerator compressor is centred around the dsPIC33EP-64MC20. It incorporates Microchip components like the MCP16331 DC/DC converter and the MIC5239 LDO voltage regulator, ensuring broad compatibility with various refrigerator compressor motors, including interior permanent magnet synchronous motors (IPMSM) and surface mount permanent magnet synchronous motors (SPMSM). The design's key features include sensorless control and a single-shunt current sensing technique, reducing bill of materials (BoM) costs. It addresses challenges in reliable compressor startup under high pressure and is engineered for low standby power consumption, making it a suitable choice for modern refrigeration systems that demand efficient and reliable compressor operation. The design features a continuous output current of 0.65Arms for up to 250W output power from a 187V to 264V single-phase input, and peaks at 350W for high-pressure operations. The system, powered by a 220V AC grid input, uses a dsPIC33EP-64MC202 DSC for motor control, with input voltage processed through filter and protection circuitry, rectified to DC, and an auxiliary power supply delivering 15- and 3.3-volt outputs to the MCU, feedback mechanisms, and a three-phase inverter bridge. The reference design is suitable for modern refrigeration systems. Microchip https://www.electronicsforu.com/electronics-projects/refrigerator-compressor-reference-design